BRAIN TOPOLOGY-DRIVEN GRAPH STRUCTURE LEARNING FOR FUNCTIONAL BRAIN NETWORK-BASED MAJOR DEPRESSIVE DISORDER DIAGNOSIS

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ABSTRACT

Resting-state functional magnetic resonance imaging is widely used to analyze abnormal brain activity in Major Depressive Disorder (MDD) by modeling the brain as a graph. Recent graph-based approaches employ graph structure learning (GSL), which adaptively infers connectivity among brain regions from data, aiming to identify subtle disease-related alterations. However, existing GSL methods rarely account for an essential characteristic of brain organization, the small-world network, which reflects local specialization and efficient global integration. Motivated by this limitation, we propose the **S**mall-world **P**attern **A**ware **R**econstructed networ**K** (SPARK), a dual-geometry GSL framework designed to incorporate this property into learned structures. SPARK first enriches node representations through biased random walks, capturing multi-scale dependencies across regions. Building on these representations, it then performs geometry-aware GSL in both Euclidean and hyperbolic spaces, enabling complementary encoding of local interactions and global modular organization. Experimental results demonstrate superior classification performance compared to existing methods.

***Index Terms*—** Graph structure learning, Small-world brain network, Dual-geometry learning, Biased random walk, Major depressive disorder

1. INTRODUCTION

Major depressive disorder (MDD) is a prevalent psychiatric condition characterized by persistent low mood, cognitive dysfunction, and impaired daily functioning [1]. Given the absence of objective biomarkers, resting-state functional magnetic resonance imaging (rs-fMRI) has emerged as a primary noninvasive tool for analyzing abnormal brain activity patterns through blood-oxygen-level-dependent (BOLD) signals [1]. To characterize these patterns, the brain can be represented as a graph where nodes correspond to regions of interest (ROIs) and edges denote functional connectivity (FC), typically defined as BOLD correlations, on which graph neural networks (GNNs) have shown strong diagnostic potential [2]. However, predefined graph structures used in conventional GNNs are non-trainable and often misaligned with downstream tasks, limiting their ability to capture subtle but disease-related patterns [2, 3, 4]. This limitation has motivated the introduction of graph structure learning (GSL) for brain disease diagnosis, which jointly infers the graph structure and learns its representations from data [3].

Recent studies have explored various GSL-based approaches, including methods that capture temporal diversity in brain activity for relational modeling, model inter-group structural variability through contrastive learning, and infer directed causal relationships among brain regions [3, 4, 5]. Although these methods have improved the flexibility of graph construction, they have yet to fully capture the intrinsic organization of the brain. The small-world topology, characterized by dense local clustering and sparse long-range connections, is a fundamental structural property of the healthy brain [6]. Incorporating this principle is thus essential for constructing biologically plausible and diagnostically relevant brain graphs. This limitation can be understood from two complementary perspectives. At the node level, most methods directly use low-order FC or raw BOLD-derived features as the basis for graph construction, thereby constraining their ability to capture high-order interactions among regions. At the structural level, current GSL approaches tend to rely solely on pairwise dependencies when constructing graphs, neglecting the small-world-like balance between locally clustered and globally integrated structures.

To address these challenges, we propose the **S**mall-world **P**attern **A**ware **R**econstructed networ**K** (SPARK), a dual-geometry GSL framework that enables brain graphs to self-organize into small-world-like structures. SPARK first enhances node representations via a biased random walk (BRW) process that propagates information across multiple steps [7], enabling the embeddings to encode multi-scale dependencies among distant regions and provide a strong foundation for structure learning. Furthermore, it performs geometry-aware GSL in both Euclidean and hyperbolic spaces. Jointly leveraging these geometries captures the distinct relational properties of each embedding space, an approach that has been shown effective for modeling complex relational structures [8, 9] yet remains unexplored in brain network modeling. Euclidean space captures locally distributed relationships among nearby regions, preserving smooth continuity, while hyperbolic space models hierarchical modular organization and emphasizes global differentiation [10]. By integrating

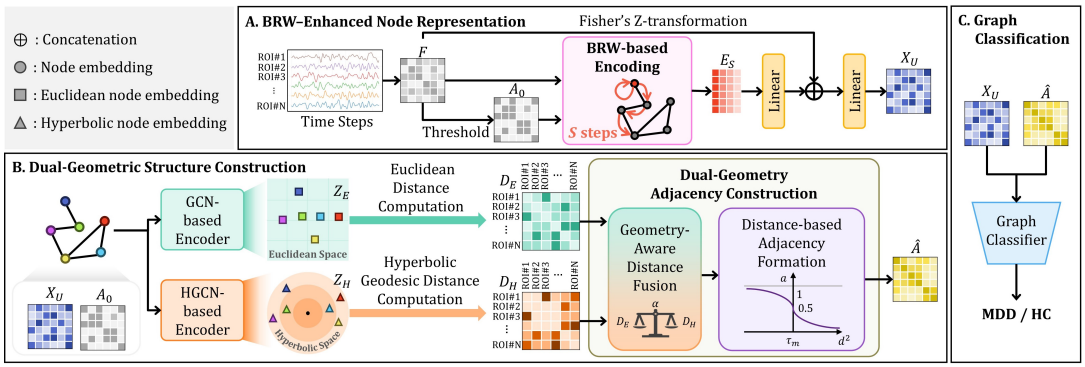

Fig. 1. An overview of the SPARK framework. The model consists of three main modules: (A) BRW-enhanced node feature representation, (B) edge construction using dual-geometry GSL, and (C) graph classification using refined graph.

these complementary geometries, SPARK reconstructs small-world-like brain networks that balance local clustering and long-range connectivity. Experiments on the REST-meta-MDD dataset demonstrate that SPARK achieves superior classification and generalization performance, revealing a biologically consistent network organization.

2. METHODS

The proposed SPARK framework consists of three stages: (1) node feature representation integrating short- and long-range topological contexts, (2) adaptive edge construction in dual geometric spaces, and (3) graph classification using the resulting features and structures, as illustrated in Fig. 1.

2.1. BRW-Enhanced Node Representation

For each subject, the brain network is modeled as an undirected graph $G = (\mathbf{X}, \mathbf{A})$, where $\mathbf{X} \in \mathbb{R}^{N\times d}$ and $\mathbf{A} \in \mathbb{R}^{N\times N}$ denote the node feature matrix and the adjacency matrix, respectively, for N ROIs. The initial node feature matrix $\mathbf{X}_0 \in \mathbb{R}^{N\times d_{\text{in}}}$ is derived from the FC matrix $\mathbf{F}$, computed using the Pearson correlation coefficient (PCC) between ROI-wise BOLD signals, followed by Fisher's z-transformation [15]. The initial adjacency matrix $\mathbf{A}_0$, serving as a warm start for structure learning, is a binary matrix obtained by retaining the top $k\%$ of the absolute FC matrix $|\mathbf{F}|$ to enforce sparsity, where k denotes the sparsity rate [16].

Each row of $\mathbf{X}_0$ indicates the short-range connectivity of a node with other ROIs. To encode long-range context, we adopt the BRW strategy from ALGA [7], which captures indirect dependencies between distant regions by simulating random walks toward strongly connected node pairs on the initial graph, as shown in Fig. 1A. For BRW, the transition matrix $\mathbf{R} \in \mathbb{R}^{N\times N}$ is computed as $\mathbf{R} = (|\mathbf{F}| \odot \mathbf{A}_0)\Delta^{-1}$, where $\odot$ denotes element-wise multiplication and Δ is the diagonal degree matrix of $|\mathbf{F}| \odot \mathbf{A}_0$. The BRW process propagates along $\mathbf{R}$ for S steps, accumulating transition probabilities across steps to capture how information flows, thereby forming long-range embeddings $\mathbf{E}_S \in \mathbb{R}^{N\times S}$. Each embedding is computed as $\mathbf{e}_i = [\mathbf{I}, \mathbf{R}, \mathbf{R}^2, \ldots, \mathbf{R}^{S-1}]_{ii} \in \mathbb{R}^S$, indicating how frequently node i is revisited during the process. The detailed formulation can be found in [7]. Since the random walk step S includes self-connections, setting $S \geq 3$ allows information to propagate beyond immediate neighbors, enabling the capture of long-range dependencies. The resulting $\mathbf{E}_S$ is linearly projected and concatenated with $\mathbf{X}_0$, followed by another linear transformation to produce the final node representation $\mathbf{X}_U \in \mathbb{R}^{N\times d}$, which encodes multi-scale contextual information for structure learning.

2.2. Dual-Geometric Structure Construction

As shown in Fig. 1B, we employ a dual-geometry GSL module that infers edges from both Euclidean and hyperbolic embeddings. The process branches into two parallel paths to compute geometry-specific distance matrices. In the Euclidean branch, the node features $\mathbf{X}_U$ are passed through a GCN layer [11] followed by a Mish activation [17] using $\mathbf{A}_0$, yielding Euclidean embeddings $\mathbf{Z}_E \in \mathbb{R}^{N\times d}$. The squared Euclidean distance between nodes i and j is computed as:

$$d_E(i,j)^2 = \|\mathbf{z}_E^i - \mathbf{z}_E^j\|_2^2. \tag{1}$$

In the hyperbolic branch, we adopt the HGCN framework [18] built on the hyperboloid model. A d-dimensional hyperbolic space with constant negative curvature $-1/K$ is defined as $\mathbb{H}^{K,d} := \{\mathbf{x} \in \mathbb{R}^{d+1} \mid \langle \mathbf{x}, \mathbf{x}\rangle_{\mathcal{L}} = -K, x_0 > 0\}$, with $K = 1$. Here, $\langle \cdot, \cdot \rangle_{\mathcal{L}}$ denotes the Lorentz inner product, defined as $\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}} = -x_0 y_0 + \sum_{i=1}^{d} x_i y_i$. To apply HGCN, the input node features $\mathbf{X}_U$ are first projected onto the tangent space at the origin, $\mathcal{T}_{\mathbf{o}}\mathbb{H}^{K,d} := \{\mathbf{v} \in \mathbb{R}^{d+1} \mid \langle \mathbf{v}, \mathbf{o}\rangle_{\mathcal{L}} = 0\}$, and then mapped onto $\mathbb{H}^{K,d}$ via the exponential map. An HGCN layer is applied to the mapped node features using $\mathbf{A}_0$, producing hyperbolic embeddings $\mathbf{Z}_H \in \mathbb{H}^{K,d}$. The detailed formulations and architectural components of HGCN can be found in [18]. We employ a single HGCN layer with the Mish

Model	ID (Site 20)					OOD (Site 20 → Site 1)		OOD (Site 20 → Site 21)	
	ACC	SEN	SPE	F1	AUC	ACC	AUC	ACC	AUC
GCN [11]	61.25±5.36	61.31±9.52	61.16±8.15	62.01±5.81	65.80±4.05	58.45±2.99	62.82±2.02	55.97±1.99	58.65±1.27
GAT [12]	57.86±2.58	60.41±3.26	55.09±3.27	59.90±2.43	58.29±4.13	59.30±2.29	61.29±1.74	55.42±1.72	56.55±2.09
GIN [13]	60.37±5.54	59.55±11.57	61.22±6.28	60.51±7.92	64.98±7.24	55.63±3.05	60.33±2.57	57.50±1.78	60.56±2.51
CRGNN [3]	55.82±7.74	58.82±11.80	52.61±9.13	57.76±8.33	54.82±8.82	53.80±3.14	54.81±2.45	51.67±1.79	50.49±2.33
ContrastPool [5]	62.78±1.47	64.43±6.18	60.95±6.61	64.21±6.13	63.76±5.19	53.80±3.14	57.24±4.46	57.64±3.10	59.94±3.64
FBNetGen [4]	64.68±2.06	68.58±4.11	60.51±7.30	66.92±1.23	68.23±3.26	54.65±3.66	58.26±2.88	56.39±1.35	58.19±2.44
BrainNetTF [14]	64.67±1.36	65.37±7.41	63.93±9.64	65.70±2.05	70.41±1.63	57.60±4.02	63.00±2.63	58.75±1.73	61.33±1.09
ALTER [7]	65.74±3.23	67.38±8.88	64.02±3.97	66.94±4.87	71.03±2.81	60.28±2.77	64.29±1.52	56.81±3.69	60.69±1.82
SPARK	**68.07±3.37**	**70.25±9.55**	**65.70±5.16**	**69.33±5.02**	**71.69±3.69**	**61.41±3.74**	**65.61±2.26**	**60.56±2.83**	**62.23±2.55**

Table 1. Comparison of classification performance in the ID and OOD settings on the REST-meta-MDD dataset. The best results are highlighted in **bold**, and the second-best are underlined.

activation function, consistent with the configuration used in the Euclidean branch. Based on the obtained embeddings $\mathbf{Z}_H$, the squared geodesic distance between nodes i and j is computed to quantify their relationship in hyperbolic space [9]:

$$d_H(i,j)^2 = K \cdot \operatorname{arcosh}^2\left(-\frac{1}{K}\langle \mathbf{z}_H^i, \mathbf{z}_H^j \rangle_{\mathcal{L}}\right). \quad (2)$$

To integrate the relational characteristics from both Euclidean and hyperbolic spaces, we fuse the distance matrices $\mathbf{D}_E$ and $\mathbf{D}_H$ using a product manifold formulation [9]. Each matrix is min–max scaled separately to align their value ranges across geometries. The unified squared distance is computed as a convex combination of the squared Euclidean and hyperbolic distances, with the learnable weighting factor $\alpha \in (0,1)$, initialized to 0.5, controlling the relative contribution of each geometry [9]. The resulting unified distance matrix $\mathbf{D}_U = [d_U(i,j)^2]$ is then transformed into a weighted adjacency matrix $\hat{\mathbf{A}}$ through a sigmoid kernel centered at the median value τ_m of $\mathbf{D}_U$, with a learnable temperature parameter $t > 0$ initialized to 1.0 [19]:

$$d_U(i,j)^2 = (1-\alpha) \cdot d_E(i,j)^2 + \alpha \cdot d_H(i,j)^2, \quad (3)$$

$$\hat{\mathbf{A}}_{ij} = \sigma\left(t \cdot (\tau_m - d_U(i,j)^2)\right). \quad (4)$$

This encourages higher edge weights between geometrically closer nodes while suppressing distant pairs. To facilitate consistent scaling across samples and stabilize training, $\hat{\mathbf{A}}$ is normalized within each sample to the $[0,1]$ range, and symmetry is enforced by reflecting the upper triangle and setting diagonal entries to one. The resulting $\hat{\mathbf{A}}$ is used as the learned graph structure for downstream classification.

2.3. Graph Classification and Optimization

The classification module, illustrated in Fig. 1C, takes the node feature matrix $\mathbf{X}_U$ (Sec. 2.1) and the learned adjacency matrix $\hat{\mathbf{A}}$ (Sec. 2.2) as inputs. The classifier consists of a two-layer GCN followed by a three-layer fully connected network with Mish activation [17] and batch normalization. The model is optimized using cross-entropy loss [4]. The code is available at https://github.com/ssurxiv/SPARK.

3. EXPERIMENTS

3.1. Experimental Setting

Datasets and Preprocessing. Experiments were conducted on the REST-meta-MDD dataset [20] using the three largest sites: Site 20 (245 MDD / 225 healthy controls (HC)), Site 1 (71 MDD / 71 HC) and Site 21 (65 MDD / 79 HC). All scans were acquired on Siemens Tim Trio 3T scanners (TR/TE = 2000/30 ms, flip angle = 90°, slice thickness/inter-slice gap = 3.0/1.0 mm, voxel size = 3.4 × 3.4 × 4.0 mm^3) and preprocessing followed the DPARSF pipeline (motion correction, normalization, smoothing, and band-pass filtering (0.01–0.08 Hz)) [16]. BOLD signals were averaged over 112 regions defined by the Harvard–Oxford Atlas [21].

Implementation Details. We evaluated performance under two settings: (1) in-distribution (ID), training and testing on Site 20 with 5-fold cross-validation; and (2) out-of-distribution (OOD), training on Site 20 and testing on Site 1 or Site 21 with five random seeds for robustness. Training employed the Adam optimizer [22] with a learning rate of 1×10^{-5}, weight decay of 1×10^{-4}, batch size of 64, and 200 epochs. Other key hyperparameters were $k = 50$, $S = 8$ and $d = 32$. Baselines followed their original configurations, using the same initial graph $G = (\mathbf{X}_0, \mathbf{A}_0)$ where applicable.

3.2. Results

Table 1 summarizes the classification results under two evaluation settings. We report the mean and standard deviation for five metrics: accuracy (ACC), sensitivity (SEN), specificity (SPE), F1-score (F1), and area under the ROC curve (AUC).

Across all metrics, SPARK achieved the highest performance among competing methods, including standard GNNs (GCN [11], GAT [12], GIN [23]), GSL-based approaches (CRGNN [3], ContrastPool [5], FBNetGen [4]), and transformer-based models (BrainNetTF [14], ALTER [7]). These results indicates its ability to capture disease-relevant structural patterns and to generalize effectively beyond the training distribution.

(a) Effect of LE and DGSL.

LE	DGSL	ACC	SEN	SPE	F1	AUC
✗	✗	61.25	61.31	61.16	62.01	65.80
✓	✗	64.00	66.53	61.27	65.69	68.22
✗	✓	63.34	67.80	59.06	65.47	68.52
✓	✓	**68.07**	**70.25**	**65.70**	**69.33**	**71.69**

(b) Effect of geometry.

Geometry	ACC	SEN	SPE	F1	AUC
Euclidean	66.77	68.61	64.77	67.99	71.16
Hyperbolic	67.41	69.44	65.24	68.75	71.02
Dual-Geometry	**68.07**	**70.25**	**65.70**	**69.33**	**71.69**

Table 2. Ablation studies on (a) LE and DGSL, and (b) embedding geometry. The best results are in **bold**, and the second-best are underlined.

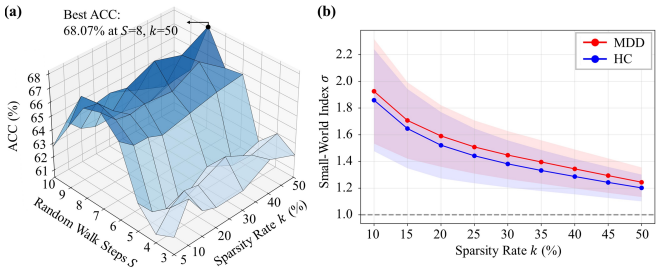

Fig. 2. (a) Classification ACC over sparsity rate k and random-walk step S in the ID setting. Darker colors denote higher ACC. (b) Small-World Index σ of learned networks for the MDD and HC groups across sparsity rates (Site 20).

3.3. Ablation Study

Ablation on LE and DGSL. Table 2(a) analyzes the contributions of the Long-range Embedding (LE) and Dual-Geometry Structure Learning (DGSL) modules. LE alone improves performance by enriching node representations with higher-order contextual information, whereas DGSL alone enhances results by adaptively refining the graph structure based on geometric distances. When combined, the model achieves the best performance across all metrics, indicating that feature-level and structure-level enhancements act synergistically to model brain connectivity. To further examine the effect of LE, Fig. 2(a) shows how model performance varies with the sparsity rate k and the number of random-walk steps S used for LE generation. Although sparsity also affects accuracy, performance generally improves as S increases, reaching a peak around $S = 8$ with $k = 50$ before declining as overly long walks introduce noise.

Analysis of Geometry-Aware Representations. Table 2(b) compares different geometric configurations for graph structure construction. The Euclidean setting builds graphs using only Euclidean embeddings, while the hyperbolic setting uses solely hyperbolic embeddings. The dual-geometry setting yields the best performance, suggesting that integrating both spaces captures complementary relational patterns. Notably, during training, the weighting factor α converged to approximately 0.5, indicating that balanced geometry contributions are optimal. This observation is further supported by Fig. 3(a), which visualizes the averaged node embeddings of the MDD and HC groups in both spaces. The Euclidean embeddings were projected onto a two-dimensional plane using PCA [24], and the hyperbolic embeddings were mapped onto the Poincaré disk [25]. In the Euclidean space, nodes are more uniformly and compactly distributed across groups. However, in the hyperbolic space, nodes exhibit clearer clustering, particularly in the occipital regions, where the MDD and HC groups show distinct spatial arrangements.

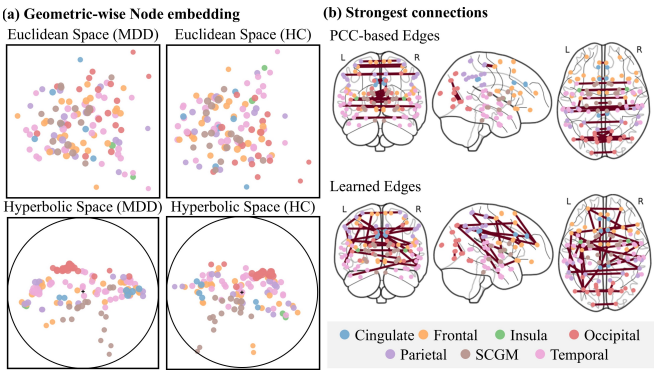

Fig. 3. Visualization of geometry-aware ROI embeddings and edge patterns (Site 20). (a) Euclidean and hyperbolic embeddings of MDD and HC groups. Larger circles indicate higher overlap. (b) Comparison of PCC-based and learned graphs.

By combining the two embeddings, we derived the learned edges and compared them with PCC-based ones in Fig. 3(b). To highlight structural differences, only the top 0.5% strongest edges are displayed. PCC-based edges mainly form symmetric connections between homologous regions, whereas learned edges emphasize inter-lobar and long-range connections, as observed in small-world organization [26]. For quantitative evaluation, we further computed the Small-World Index σ [6] of the learned networks by retaining the top-k% strongest edges, as shown in Fig. 2(b). Across varying sparsity rates k, the learned structures consistently satisfy the small-world criterion ($\sigma \geq 1.0$) [6, 26], with group differences observed.

4. CONCLUSION

This paper presents SPARK, a GSL framework designed to model brain networks with small-world properties through DGSL leveraging BRW-enhanced node features. Experimental results demonstrate that SPARK achieves superior classification and generalization performance while revealing biologically consistent small-world organization, underscoring its promise for interpretable diagnosis of MDD.

5. ACKNOWLEDGEMENT

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